Credit Card Default Prediction Based on Machine Learning

Abstract: With the increasing importance of credit risk management in financial institutions, accurate default prediction models are essential for minimizing losses and optimizing lending strategies. This study employs multiple machine learning algorithms, including logistic regression, random forests, gradient boosting, XGBoost, and so on, to predict whether a credit card holder will default in the upcoming month. Through systematic data preprocessing, feature engineering, and model optimization, the gradient boosting model emerged as the most effective predictor, achieving an AUC-ROC score of 0.7801 on the test set. The analysis identifies key risk factors, with the most recent payment status (PAY_1) being the most significant predictor of default. This research provides practical insights for financial institutions seeking to implement data-driven risk management systems.

Keywords: Credit Risk, Default Prediction, Machine Learning, Gradient Boosting

1 Introduction

1.1 Background and Significance

Credit card lending represents a significant component of consumer finance worldwide, with outstanding credit card debt reaching trillions of dollars globally. The 2008 financial crisis highlighted the critical importance of accurate risk assessment in lending practices. Traditional credit scoring models, such as FICO scores, while valuable, often fail to capture nuanced patterns in payment behavior and may not adapt quickly to changing economic conditions [1]. Machine learning approaches offer the potential to improve prediction accuracy by identifying complex patterns in historical data and incorporating a wider range of features than traditional models [2].

The consequences of credit card default extend beyond individual financial institutions to affect the broader economy through increased borrowing costs, reduced credit availability, and potential systemic risks. Effective default prediction models enable banks to implement proactive measures, such as adjusting credit limits, offering payment plans, or targeting interventions to high-risk customers before default occurs.

1.2 Research Objectives

This project aims to address three key questions in credit risk prediction.

(1) How do different machine learning algorithms compare in predicting default risk?

(2) What is the optimal machine learning model for this prediction task?

(3) Which features are most predictive of credit card default?

2 Data Acquisition and Description

2.1 Dataset Source and Characteristics

The data is sourced from the " Credit Default Prediction Challenge | Kaggle" competition [1]on Kaggle. Based on data availability, the "NEK 310 Lecture 11 | Kaggle" dataset[2], which has the same data but is publicly available, was selected for download.

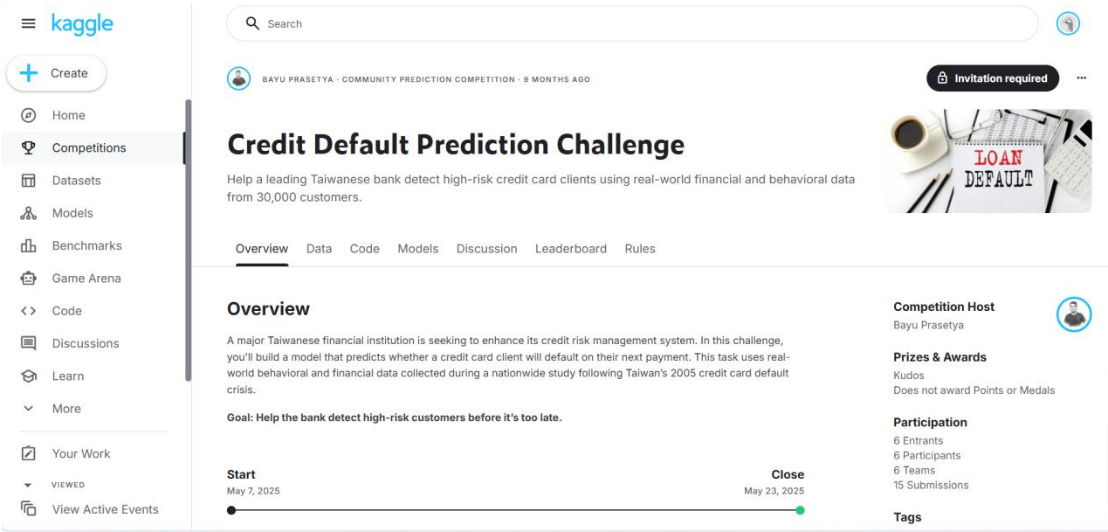

Fig.1 Screenshot of the Credit Default Prediction Challenge competition

The complete dataset includes 30,000 observations with 25 variables. Each observation represents an individual credit card holder at a specific point in time.

2.2 Variable Description

The dependent variable of this dataset is a categorical variable (whether to default next month), which indicates that this is a classification prediction problem. The independent variables include three main categories of variables: Demographic Variables(SEX, $EDUCATION$, $MARRIAGE$, AGE), Historical Payment Variables(PAY_1 to PAY_6), Financial Transaction Variables($BILL\_ATM1$ to $BILL\_ATM6$ and PAY_ATM1 to PAY_ATM6).

The key descriptions of these variables are listed in Table 1.

Table 1 Variable description

Variable Name	Description
ID	Unique identifier for each client
$LIMIT\_BAL$	Credit limit given to the client (NT dollars)
SEX	Gender (1 = male, 2 = female)
$EDUCATION$	Education level (1 = graduate, 2 = university, 3 = high school, 4 = others)
$MARRIAGE$	Marital status (1 = married, 2 = single, 3 = others)
AGE	Age of the client (in years)
PAY_0 to PAY_6	Payment status from April to September 2005. (0 = on time, -1 = pay duly, 1–9 = delay in months)

[1] 网址：https://www.kaggle.com/competitions/credit-default-prediction-challenge/overview
[2] 网址：https://www.kaggle.com/competitions/nek-310-lecture-11/overview

Continued Table 1

Variable Name	Description
$BILL\_AMT1$ to $BILL\_AMT6$	Amount of bill statement over the past 6 months
PAY_AMT1 to PAY_AMT6	Amount paid by the client in the past 6 months
$default\_payment\_next\_month$	Target variable: 1 = default next month, 0 = no default

2.3 Data Quality Assessment

Initial data quality checks revealed several important characteristics.

(1) Missing Values: The dataset contains no missing values.

(2) Data Consistency: All variables maintain consistent formats across observations.

(3) Outliers: Some financial indicators exhibit extreme values, which require careful handling. However, in the current situation, I believe the range of numerical data is normal, but I have also identified some outliers in categorical variables (which will be addressed later)

(4) Class Imbalance: The default rate is 22.12%, representing 6,636 default cases out of 30,000 total observations. This conforms to the characteristics of credit default events and has an appropriate sample size.

In summary, this dataset is basically clean, but the categorical variables need to be processed.

3 Data Preprocessing and Feature Engineering

3.1 Data Cleaning

The preprocessing phase addressed several data quality issues:(1) Extreme values in financial variables were analyzed but retained, as they represent legitimate variations in customer behavior rather than data errors; (2) Educational Variable Recoding: Values 0, 5, and 6 in the EDUCATION variable were recoded as 4 (others) to align with the defined categories and reduce noise; (3) Marital Status Correction: The value 0 in MARRIAGE was recoded as 3 (others) for consistency, ensuring all values fell within the defined range (1-3).

3.2 Feature Engineering

New features were created to capture patterns not explicit in the original variables, increasing the feature count from 23 to 33.

The key descriptions of these variables are listed in Table 2.

Table 2 Description of Feature Engineering Variables

Aspects	Variable Name	Description
Payment Behavior Features	PAY_MEAN	Average payment delay across six months
	PAY_MAX	Maximum payment delay observed in past six months
	PAY_STD	Standard deviation of payment delays (measures payment consistency)

Continued Table 2

Aspects	Variable Name	Description
Financial Pattern Features	*BILL_MEAN*	Average monthly bill amount over six months
	BILL_MAX	Maximum monthly bill amount over six months
	BILL_TOTAL	Total billed amount over six months
	PAY_AMT_MEAN	Average monthly payment amount over six months
	PAY_AMT_TOTAL	Total payments made over six months
Derived Risk Indicators	*CREDIT_UTILIZATION*	Ratio of most recent bill to credit limit (measures credit usage intensity)
	AGE_GROUP	Categorized age into five groups (20-30, 30-40, 40-50, 50-60, 60+)

4 Exploratory Data Analysis

4.1 Overall Data Characteristics

The dataset comprises 30,000 credit card holders with an average credit limit of 167,484 NTD. The average age is 35.5 years, with a standard deviation of 9.22 years, indicating a relatively young customer base. Gender distribution shows 60.4% female and 39.6% male customers, while educational attainment is concentrated at the university level (46.8%).

4.2 Target Variable Analysis

The default rate in the dataset is 22.12%, representing 6,636 default cases out of 30,000 total observations.

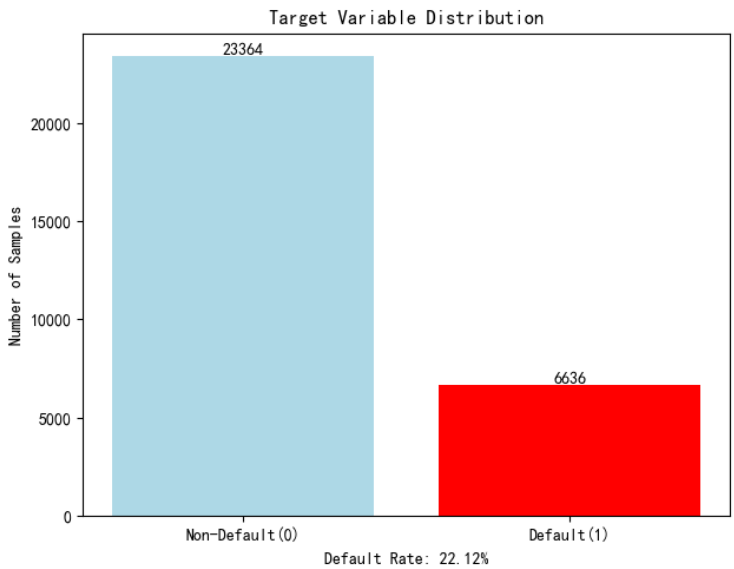

Fig.2 Target Variable Distribution

4.3 Feature Relationships with Default

Figure 3 shows the default rates of some key features.

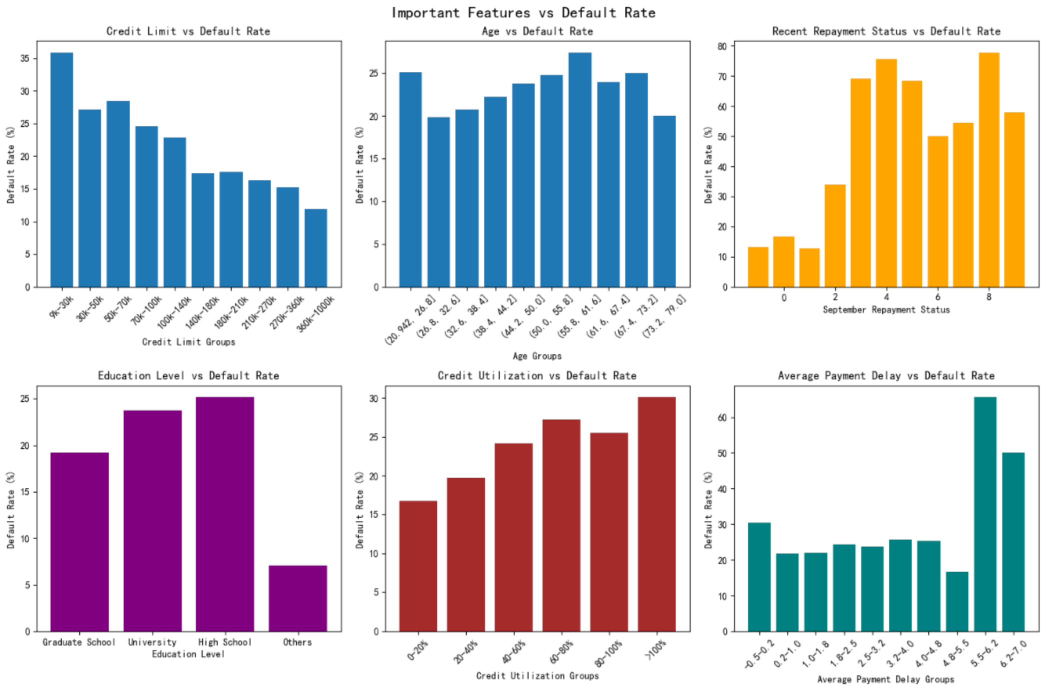

Fig.3 Default rates in different groups

4.4 Correlation Analysis

A correlation matrix reveals that payment status variables show the highest correlations with default (0.187 to 0.325). Meanwhile, engineered features like PAY_STD and PAY_AMT_MEAN show moderate correlations (0.215 and 0.164 respectively).Financial amount variables show relatively lower direct correlations with default, and demographic features show weaker correlations.

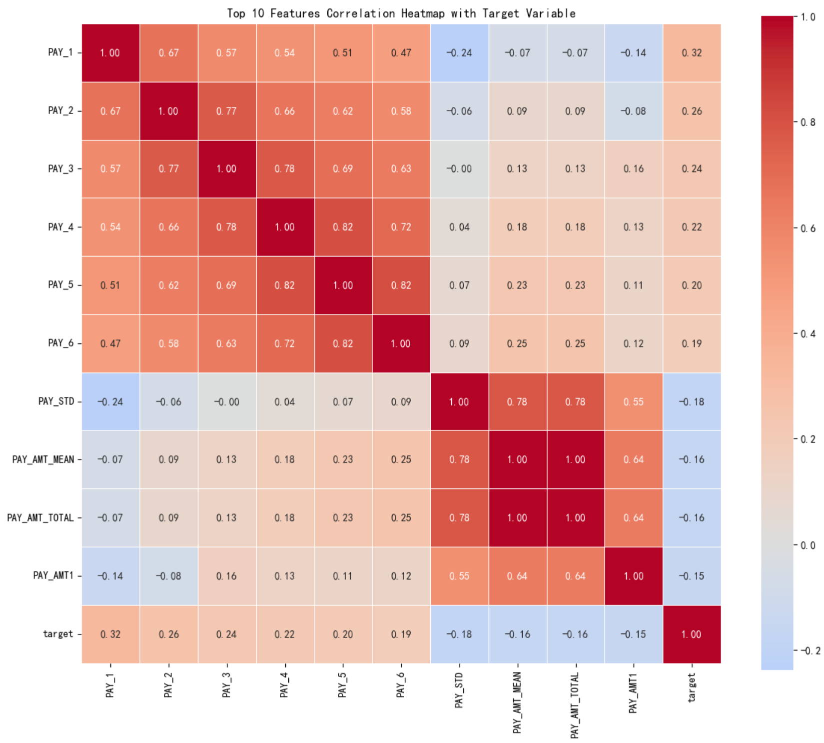

Fig.4 Correlation between important features and the target variable

5 Credit Card Default Prediction Model

5.1 Data Processing

(1) Scaling: All numerical features were standardized to mean=0 and variance=1 to ensure equal contribution from features with different scales.

(2) Encoding: Ensure all categorical variables (SEX, EDUCATION, MARRIAGE, AGE_GROUP) were one-hot encoded, expanding the feature set to 43 variables.

(3) Train-Validation-Test Split: Data were divided into training (60%, 18,000 samples), validation (20%, 6,000 samples), and test (20%, 6,000 samples) sets using stratified sampling to maintain the 22.12% default rate across all splits.

5.2 Selected Algorithms

This project selected a total of nine algorithms belonging to three major categories.

5.2.1 Logistic Regression

As a baseline model, logistic regression provides interpretable coefficients and establishes basic expected performance [3].

5.2.2 Tree-Based Models

(1) Decision Tree: A simple tree model provides interpretable rules but tends to overfit [4].

(2) Random Forest: An ensemble of decision trees that reduces variance through bootstrap aggregation and feature randomization [5].

(3) Gradient Boosting: Sequentially builds trees that correct previous errors, achieving the best performance [6].

(4) XGBoost: An optimized gradient boosting implementation [7].

(5) LightGBM: A gradient boosting framework using histogram-based algorithms for faster training [8].

5.2.3 Other Classifiers

(1) Support Vector Machine (SVM): Finds optimal hyperplane separating classes [9].

(2) K-Nearest Neighbors (KNN): Classifies based on similarity to nearest training examples [10].

(3) Naive Bayes: Applies Bayes theorem with strong independence assumptions.

5.3 Evaluation Metrics

Multiple evaluation metrics provide complementary perspectives. (1) Accuracy: Overall correct prediction rate; (2) Precision: Proportion of predicted defaults that are actual defaults; (3) Recall: Proportion of actual defaults correctly predicted; (4) F1-Score: Harmonic mean of precision and recall; (5) AUC-ROC: Area under the Receiver Operating Characteristic curve.

6 Results and Model Comparison

6.1 Initial Model Performance

The performance of all models on the validation set shows clear differences in predictive capability.

Table 3 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Gradient Boosting	0.8215*	0.6834*	0.3595	0.4711	0.7791*
LightGBM	0.8190	0.6676	0.3617	0.4692	0.7759
XGBoost	0.8118	0.6296	0.3625	0.4601	0.7647
Random Forest	0.8128	0.6386	0.3542	0.4556	0.7634
Logistic Regression	0.7413	0.4398	0.6194*	0.5144*	0.7456
Naive Bayes	0.7363	0.4214	0.5154	0.4637	0.7229
SVM	0.8177	0.6795	0.3323	0.4464	0.7199
Decision Tree	0.8020	0.5879	0.3504	0.4391	0.7096
K-Nearest Neighbors	0.7835	0.5183	0.2992	0.3794	0.6897

The ones marked with "*" in the table are the models with the best performance for each indicator.

6.2 Model Selection

As shown in Table 3, both the gradient boosting model and the logistic regression model perform well, and each has its own strengths.

We can select the appropriate model based on the bank's business preferences. For a profit-oriented model (aiming to increase the number of good customers while accurately predicting defaults): choose Gradient Boosting (with high Precision and AUC-ROC score). For a risk-averse model (preferring to err on the side of caution rather than letting a bad case go unaddressed): choose Logistic Regression (with high Recall).

Due to the length limitation of the project, only the gradient boosting model will be selected for use here.

7 Model Optimization

7.1 Gradient Boosting Hyperparameter Tuning

Given gradient boosting's superior performance, extensive hyperparameter optimization was conducted using randomized search with 5-fold cross-validation. The search explored 50 parameter combinations, with the optimal parameters have been determined as shown in Table 4.

Table 4 Gradient Boosting Model Hyperparameters

Parameter	Value	Description
subsample	0.7	Fraction of samples used for fitting individual base learners
n_estimators	50	Number of boosting stages (trees) to be built
min_samples_split	2	Minimum number of samples required to split an internal node

Continued Table 4

Parameter	Value	Description
min_samples_leaf	1	Minimum number of samples required to be at a leaf node
max_features	0.7	Fraction of features to consider when looking for the best split
max_depth	4	Maximum depth of individual regression estimators (trees)
loss	'exponential'	Loss function to be optimized
learning_rate	0.1	Shrinks the contribution of each tree (step size)
criterion	'squared_error'	Function to measure the quality of a split

7.2 Optimization Results

The optimization process improved the cross-validation AUC from 0.7791 to 0.7801, representing a 0.0009 improvement. It also raised the Precision from 0.6834 to 0.7003, an increase of 0.0170.

7.3 Performance Comparison

The optimized model showed mixed improvements on the validation set.

Table 5 Model Performance: Before vs After Hyperparameter Tuning

Metric	Before Tuning	After Tuning	Improvement
Accuracy	0.8215	0.8192	-0.0023
Precision	0.6834	0.7003	+0.0170
Recall	0.3595	0.3188	-0.0407
F1-Score	0.4711	0.4381	-0.0330
AUC-ROC	0.7791	0.7801	+0.0009

While AUC-ROC improved slightly, the decrease in recall suggests the optimized model became more conservative in predicting defaults, prioritizing precision over recall. This trade-off may be acceptable depending on the business context (build a profit-oriented model aiming to increase the number of good customers while accurately predicting defaults).

8 Feature Importance Analysis

8.1 Overall Feature Importance

The gradient boosting model's feature importance analysis reveals that recent payment behavior dominates default prediction.

Table 6 Feature Importance Analysis

Feature	Importance	Category	Interpretation
PAY_1	47.6%	Payment Behavior	Most recent payment status - strongest predictor
PAY_2	4.6%	Payment Behavior	Payment status from two months ago

Continued Table 6

Feature	Importance	Category	Interpretation
PAY_STD	6.6%	Payment Consistency	Standard deviation of payment delays
CREDIT_UTILIZATION	4.6%	Credit Usage	Ratio of used credit to available limit
LIMIT_BAL	3.5%	Credit Profile	Total credit limit amount

Cumulatively, the top 5 features account for 73.5% of the predictive power, while the top 10 features reach 84.8%. The first 18 features achieve 95% cumulative importance, suggesting that a relatively focused feature set could achieve near-optimal performance.

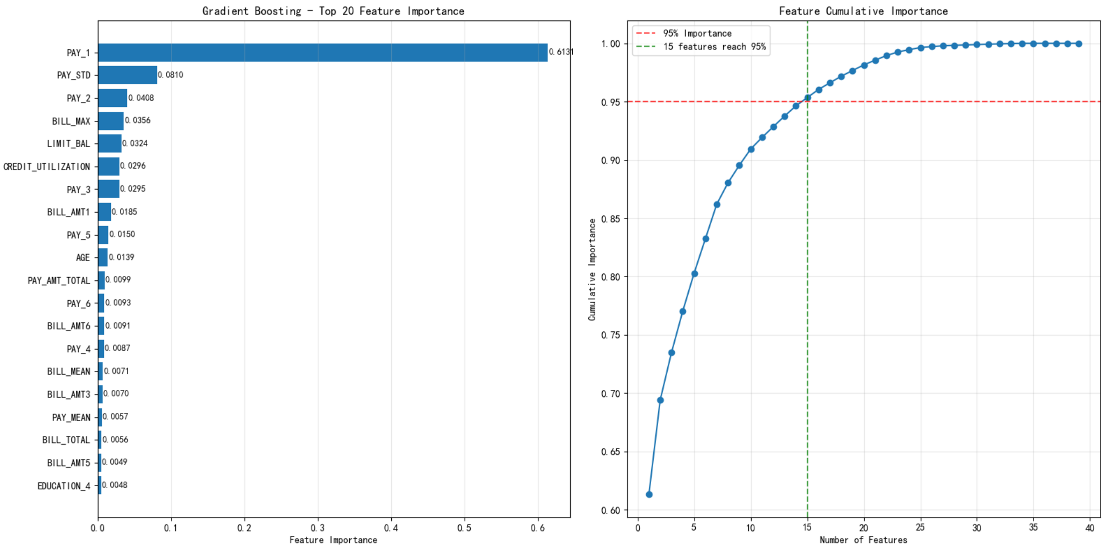

Fig.5 Feature Importance Analysis

8.2 Feature Category Analysis

Grouping features by category provides additional insights:(1) Payment Behavior Features: 7 features, 67.3% of total importance; (2) Financial Transaction Features: 8 features, 13.9% of total importance; (3) Engineered Features: 5 features, 11.2% of total importance; (4) Demographic Features: 2 features, 4.9% of total importance; (5) Repayment Amount Features: 2 features, 2.7% of total importance

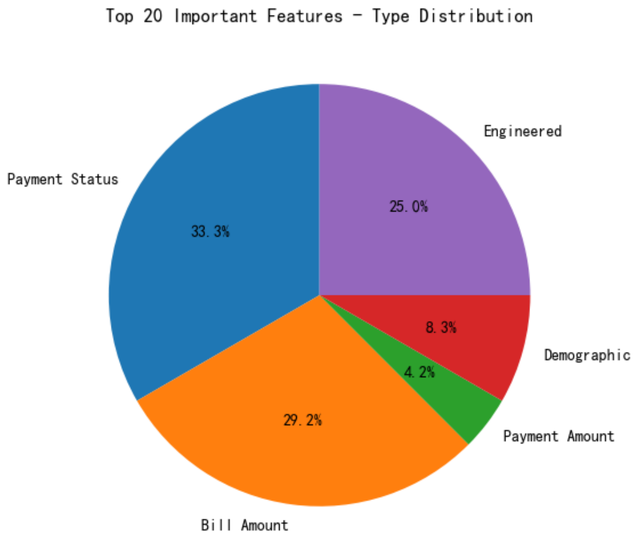

Fig.6 The importance of different feature categories

The dominance of payment behavior features (67.3%) aligns with financial intuition—past payment behavior is the best predictor of future behavior. However, the significant contribution of engineered features (11.2%), particularly credit utilization and payment statistics, demonstrates the value of feature engineering.

8.3 Business Interpretation

The feature importance results have clear business implications.

(1) Recent Payment Monitoring. Since PAY_1 accounts for nearly half of predictive power, banks should prioritize monitoring and intervening when customers show recent payment delays.

(2) Payment Pattern Analysis. The importance of PAY_STD (6.6%) indicates that inconsistent payment behavior is a strong risk indicator, possibly more telling than occasional late payments.

(3) Credit Management. The combined importance of LIMIT_BAL (3.5%) and CREDIT_UTILIZATION (4.6%) suggests that both absolute credit amount and relative usage matter for risk assessment.

(4) Model Simplification Potential. With 18 features capturing 95% of predictive power, simpler models focusing on key features could achieve nearly the same performance with better interpretability.

9 Final Model Evaluation

The optimized gradient boosting model achieved the following performance on the held-out test set.

Table 7 Model Performance on test set

Metric	Value	Interpretation
Accuracy	0.8187	The model correctly predicts 81.87% of all cases
Precision	0.6700	When the model predicts positive, it's correct 67.00% of the time
Recall	0.3549	The model identifies 35.49% of all actual positive cases

Continued Table 7

Metric	Value	Interpretation
F1-Score	0.4640	Harmonic mean of Precision and Recall (balanced measure)
AUC-ROC	0.7807	Model's ability to distinguish between classes is 78.07%

The consistency between validation (AUC-ROC: 0.7807) and test performance (AUC-ROC: 0.7801) indicates good generalization without overfitting.

10 Conclusion

This project successfully developed a machine learning framework for credit card default prediction, achieving an AUC-ROC of 0.7801 using an optimized gradient boosting model.

Two models that performed exceptionally well in the prediction task have been identified, each with its own advantages. For profit-oriented banks, the gradient boosting method can be selected, as it outperforms traditional models in precise prediction tasks. For risk-averse banks, the logistic regression model can be used to avoid risk customers.

Recent payment behavior is the strongest predictor of future default, with PAY_1 alone accounting for 47.6% of feature importance. Meanwhile, Feature engineering creates valuable predictors not present in the original data, with engineered features contributing 11.2% of total importance.

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